

Cyber Security & Ethical Hacking Course Outline

Week 1-2: Introduction to Cyber Security

- **Week 1: Basics of Cyber Security**
 - Importance of Cyber Security
 - Types of Attacks: Phishing, Malware, DDoS, etc.
 - CIA Triad (Confidentiality, Integrity, Availability)
 - Cyber Security Terminologies and Concepts
 - Introduction to the OSI Model and Network Protocols
- **Week 2: Understanding the Threat Landscape**
 - Common Cyber Threats and Vulnerabilities
 - Current Trends in Cyber Attacks
 - Understanding Attackers: Motives, Methods, and Techniques
 - Risk Management and Security Policies

Week 3-4: Networking and System Security

- **Week 3: Networking Basics for Ethical Hacking**
 - IP Addressing and Subnetting
 - TCP/IP and Common Network Protocols
 - Understanding Firewalls, VPNs, and Proxies
 - Network Vulnerabilities and Defense Mechanisms
- **Week 4: System Security**
 - Understanding Operating Systems (Windows, Linux)
 - Security Configurations for Servers and Workstations
 - Common System Vulnerabilities and Misconfigurations
 - Hardening Operating Systems

Week 5-6: Ethical Hacking Foundations

- **Week 5: Introduction to Ethical Hacking**
 - Ethical Hacking vs. Malicious Hacking
 - The Legal Framework of Ethical Hacking (Laws and Regulations)
 - Phases of Hacking (Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks)
 - Setting Up Your Hacking Lab (Kali Linux, Virtual Machines, etc.)
- **Week 6: Footprinting and Reconnaissance**
 - Information Gathering Techniques
 - Open Source Intelligence (OSINT) Tools
 - Using WHOIS, Nslookup, Shodan, etc.
 - Passive vs. Active Reconnaissance

Week 7-8: Vulnerability Analysis and Scanning

- **Week 7: Vulnerability Analysis**
 - Types of Vulnerabilities (Software, Network, Physical)
 - Vulnerability Assessment Tools (Nessus, OpenVAS, Nexpose)
 - Identifying and Prioritizing Vulnerabilities
 - Reporting Vulnerabilities
- **Week 8: Scanning and Enumeration**
 - Network Scanning Techniques (Nmap, Netcat)
 - Port Scanning, Ping Sweeps, and Service Detection
 - Identifying Open Ports and Running Services
 - Enumerating Users, Services, and Shares

Week 9-10: Exploitation and Web Application Security

- **Week 9: Exploiting Vulnerabilities**
 - Introduction to Exploitation Techniques
 - Exploiting System and Network Vulnerabilities
 - Buffer Overflows, SQL Injection, and Cross-Site Scripting (XSS)

- **Week 10: Web Application Security**

- Common Web Application Attacks (SQL Injection, XSS, CSRF)
- OWASP Top 10 Vulnerabilities
- Web Application Security Testing Tools (Burp Suite, OWASP ZAP)
- Securing Web Applications

Week 11-12: Post-Exploitation and Incident Response

- **Week 11: Post-Exploitation Techniques**

- Maintaining Access and Privilege Escalation
- Clearing Logs and Covering Tracks
- Lateral Movement within a Network
- Exfiltration of Data

- **Week 12: Incident Response and Countermeasures**

- Incident Detection and Response (IDR)
- Creating Incident Response Plans
- Digital Forensics and Evidence Collection
- Implementing Security Controls (IDS/IPS, Antivirus, SIEM)

Final Week: Capstone Project and Certification Preparation

- **Week 13: Capstone Project**

- Students will perform a full ethical hacking cycle on a simulated environment, including reconnaissance, vulnerability scanning, exploitation, and reporting.
- Write a detailed report of findings and recommendations for securing the system.

- **Week 14: Certification Preparation and Exam**

- Review of Core Topics
- Mock Tests
- Final Exams